# CS 374A Review MT2

## Electric Boogaloo

ACM @ UIUC

November 7, 2026



# Disclaimers and Logistics

- This review session is being recorded. Recordings and slides will be distributed on acadaemic.acm.illinois.edu after the end.
- **Agenda:** We'll quickly review all topics likely to be covered, then go through a practice exam, then review individual topics by request.
- Please let us know if we're going too fast/slow, not speaking loud enough/speaking too loud, etc.
- If you have a question anytime during the review session, please ask! Someone else almost surely has a similar question.
- We'll provide a feedback form at the end of the session.

# Recursion

- **Definition:** Reducing the problem to a smaller instance of itself, where eventually we can terminate in a base case.
  - Think: If we have a problem of size $n$, we want to continuously reduce to a problem smaller than $n$.
  - Example: Binary search, Karatsuba

**Template**

```
1: procedure AMAZINGRECURSIVEALGO(n)
2:     if n == [some base case] then
3:         return [value]
4:     else
5:         return AmazingRecursiveAlgo(n/2)
```

if $n \leq 1000$
Brute Force

- Similar to **induction**!

# Recursion: Runtime Analysis

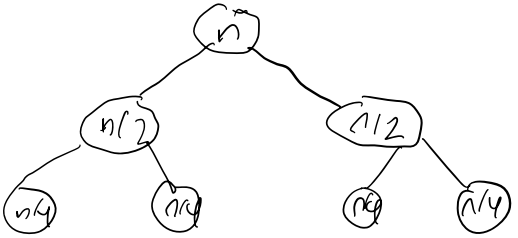
- **General Form:**

$$T(n) = \underbrace{r}_{\text{\# of subproblems}} \cdot \overbrace{T\left(\frac{n}{c}\right)}^{\text{work at each subproblem}} + \underbrace{f(n)}_{\text{work at current level}}$$

  - Describes how the amount of work changes between each level of recursion.
  - We can solve for a **time complexity** that describes the scaling behavior of the algorithm at hand.

Merge Sort: $T(n) = 2T\left(\frac{n}{2}\right) + O(n) \rightarrow O(n \log n)$

$T(n) =$



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$$\begin{aligned}
\text{Decreasing: } r \cdot f(n/c) = \kappa \cdot f(n) \text{ where } \kappa < 1 &\implies T(n) = O(f(n)) \\
\text{Equal: } r \cdot f(n/c) = \kappa \cdot f(n) \text{ where } \kappa = 1 &\implies T(n) = O(f(n) \cdot log_c n) \\
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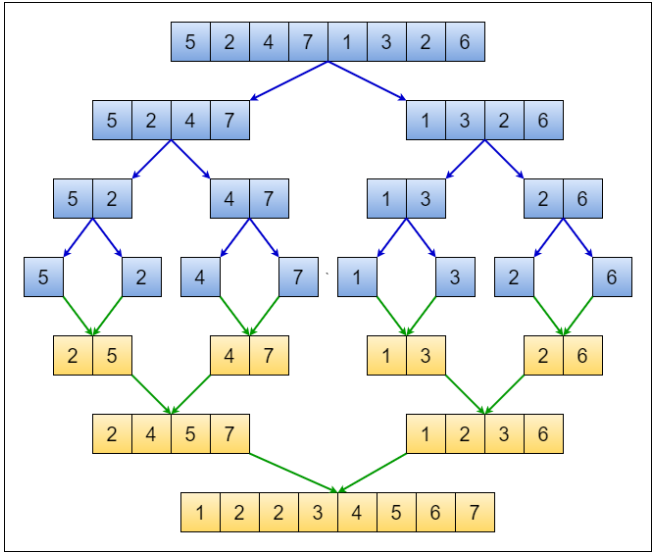
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  - **Intuition:** If each level contains more work than the level below it, then the root level will dominate. If each level contains the same amount of work, then we have $log_c n$ levels with $f(n)$ work. If each level contains less work than the work below it, then the leaf nodes will dominate.

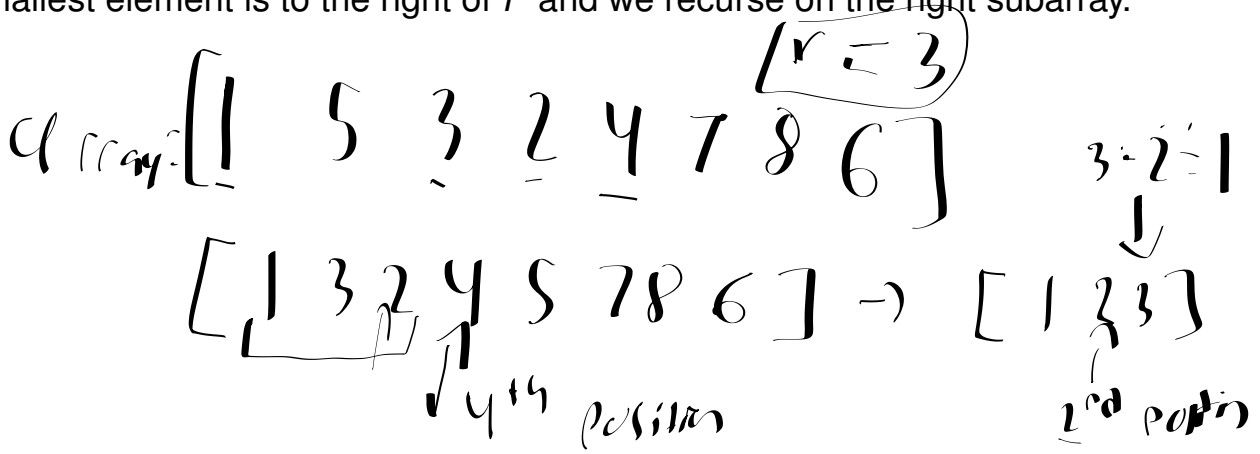
# Divide and Conquer Algos: Merge Sort

- **Purpose:** Sort an arbitrary array.
- **Time Complexity**: $O(n \log n)$
- **Intuition:** Three phases: (a) split the array in half, (b) sort each side, (c) merge the sorted halves by repeatedly comparing smallest elements on each side not yet inserted.



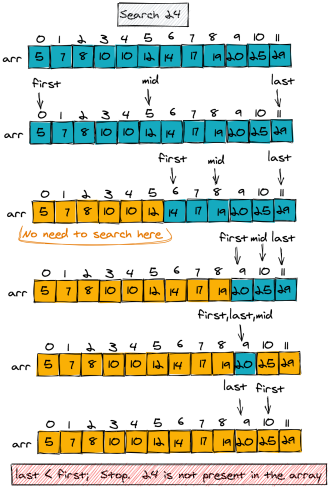
# Divide and Conquer Algos: MomSelect

- **Purpose:** Get the $r^{th}$ smallest element in an arbitrary array.
- **Time Complexity**: $O(n)$
- **Intuition**: Pick a pivot $P$ with a value $P_V$ and rearrange the array such that all the elements that are less than $P_V$ are to the left of $P$ and all the elements that are greater than $P_V$ are to the right of P. If the length of the array of elements that are less than $P_V$ is greater than $r$, then we know that the $r^{th}$ smallest element is to the left of $P$ and we recurse on the left subarray. Otherwise, we know that the $r^{th}$ smallest element is to the right of $P$ and we recurse on the right subarray.



# Divide and Conquer Algos: Binary Search

- **Purpose:** Find the existence of an element in a sorted array
- **Time Complexity**: $O(\log n)$
- **Intuition**: Say we are trying to find the value $k$. Pick the middle element $M$ in the array. If $k > M$, the element must be to the right of $M$ and we recurse on the right. Otherwise, we recurse on the left.



# Divide and Conquer Algos: Karatsuba's

- **Purpose:** Multiplication
- **Time Complexity**: $O(n^{\log_2 3}) \approx O(n^{1.585})$
- 3 Phases:
  - **Divide**: Represent $x$ as $x_0 p^n + x_1$, $y$ as $y_0 p^n + y_1$
  - **Recurse**: Calculate $x_0 y_0$, $x_1 y_1$, $(x_0 + x_1)(y_0 + y_1)$
  - **Combine**: Use the three results to calculate our final answer.

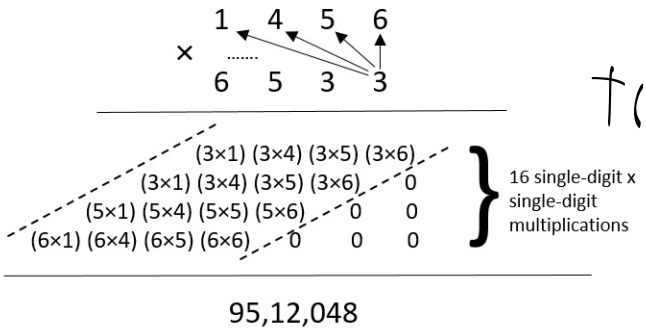
$x = 1234$

$y = 5678$

$x = 12 \cdot 10^2 + 34$

$y = 56 \cdot 10^2 + 78$

$T(n) = 3T(\frac{n}{2}) + c(n)$



# Dynamic Programming

- It's backtracking, but we compute all of the subproblems iteratively.
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  1. English Description
  2. Recurrence
  3. Memoization Structure
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  6. Runtime

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  6. Runtime
- **How to solve a DP:**
  - Identify how we can take advantage of a recursive call on a smaller subset of the input space.
  - Identity base cases
  - Identity recurrences (they should cover all possible cases at each step)

# Dynamic Programming: Longest Increasing Subsequence

1. **English Description:** $\text{LIS}(i, j) =$ the length of the longest increasing subsequence in $A[i..n]$, such that every element in the subsequence is $> A[j]$. Here, $j = 0$ can be used as a sentinel with $A[0] = -\infty$.
2. **Recurrence:**

$$\text{LIS}(i, j) = \begin{cases} 0 & \text{if } i > n, \\ \text{LIS}(i+1, j) & \text{if } A[i] \leq A[j], \\ \max\left(\text{LIS}(i+1, j),\ 1 + \text{LIS}(i+1, i)\right) & \text{if } A[i] > A[j]. \end{cases}$$

3. **Memoization Structure:** A 2D array $LIS[0..n, 0..n]$, storing all subproblem solutions $\text{LIS}(i, j)$.
4. **Solution Location:** The length of the LIS of the entire array is found at $\text{LIS}(1, 0)$, starting from the first element with the sentinel.
5. **Evaluation Order:** Fill the table in decreasing order of $i$.
6. **Runtime:** The table has $O(n^2)$ entries, each computed in $O(1)$, so the total runtime is $O(n^2)$.

# Graphs

- **Definition:** A set of vertices $V$ connected by a set of edges $E$. Individual edges are notated as $(u, v)$, where $u, v \in V$.
  - They are usually represented as **adjacency lists** or **adjacency matrices**



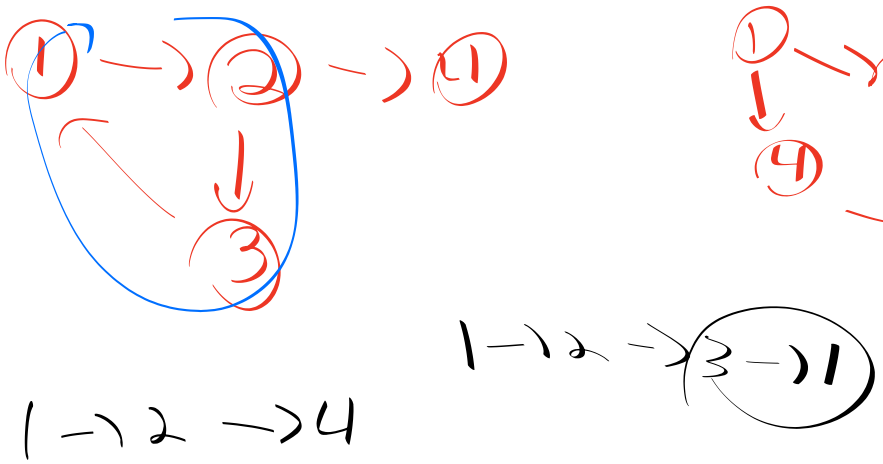
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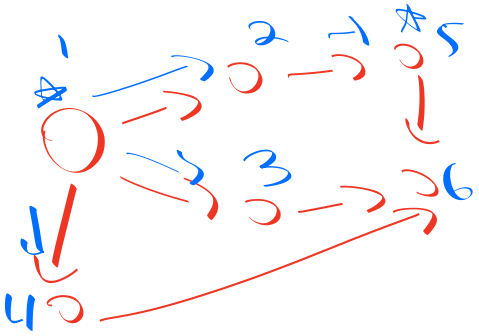
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- **Cycle:** A sequence of distinct vertices where each pair of consecutive vertices have an edge **and** the first and last vertices are connected.

# Graph Algorithms: Traversal

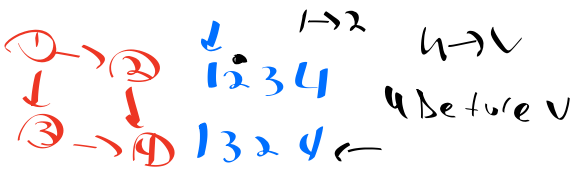
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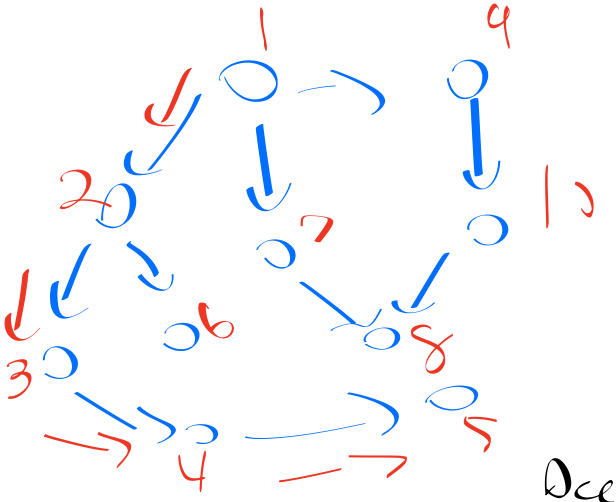
- **BFS:**
  - **Purpose:** Reachability, Shortest Path (unweighted graph)
  - **Implementation details:** Add your neighbours to a **queue**, pop from the queue to get next node
  - **Runtime:** $O(V + E)$



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- **DFS:**
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  - **Runtime:** $O(V + E)$

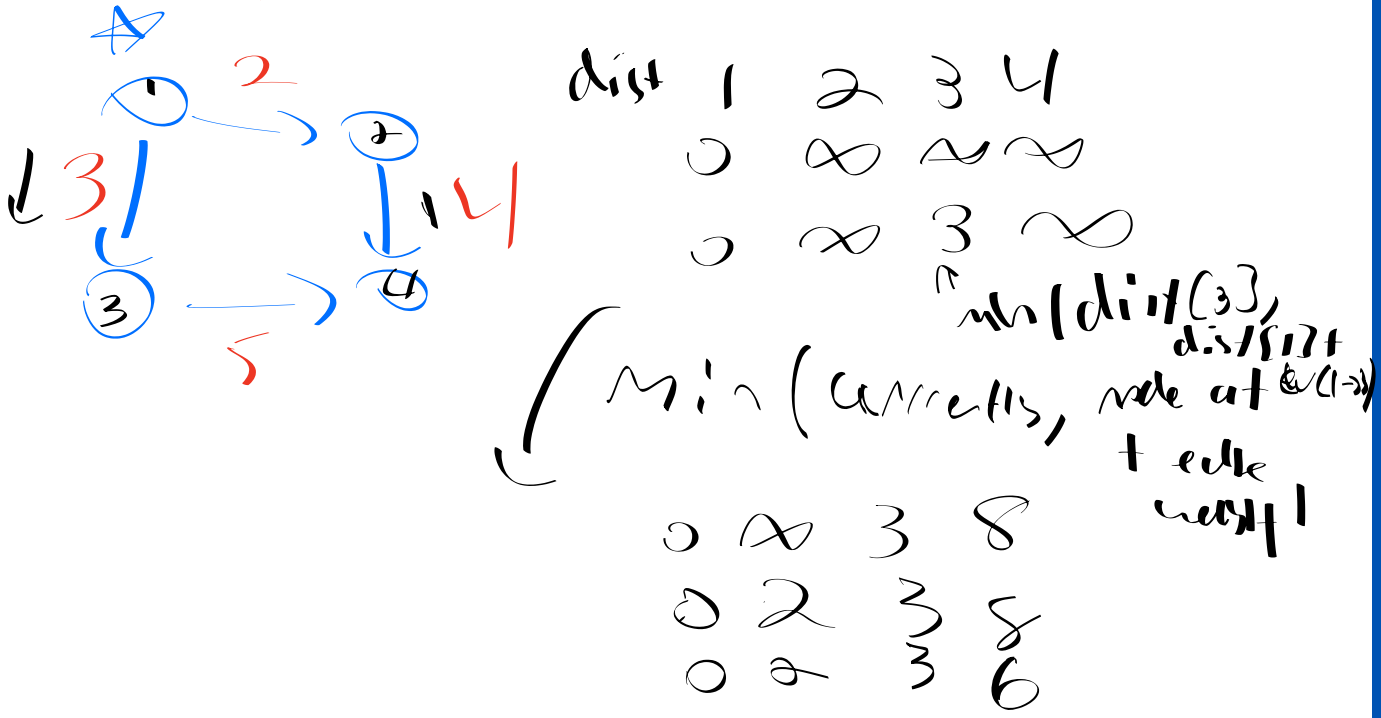






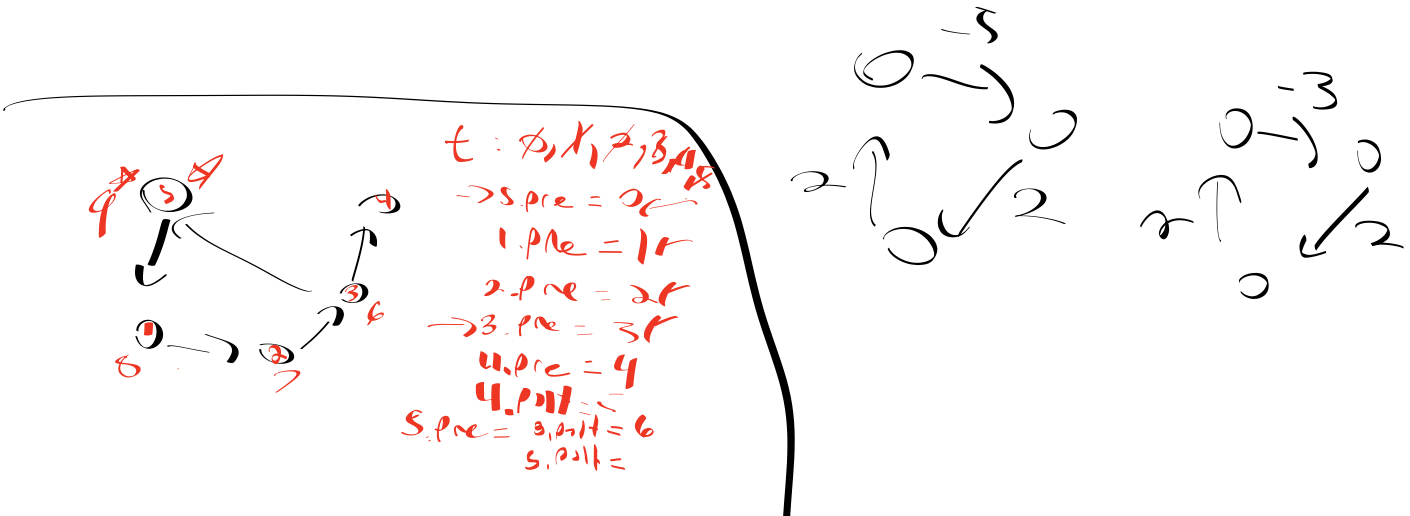
# Graph Algorithms: Shortest Path

- **Dijkstra's**
  - **Purpose:** SSSP, no negative edges
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  - **Runtime:** $O(m \log n)$ (with **Fibonacci Heap**, $O(m + n \log n)$)



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- **Bellman-Ford:**
  - **Purpose:** SSSP, yes negative weights. Will detect negative cycles.
  - **Implementation:** Dynamic Programming recurrence:
    - $d(v, k)$ is the shortest-walk distance from $s$ to $v$ using at most $k$ edges
    - $d(v, k) = \min\left(d(v, k-1), \min_{u \to v} d(u, k-1) + \ell(u \to v)\right)$
  - **Runtime:** $O(mn)$



# Graph Algorithms: Longest Path

## Longest path in a Directed Acyclic Graph (DAG)

- **Purpose:** To find the longest simple path (no repeating vertices) by weight in a graph which is guaranteed to be a DAG[1].
- **Runtime**[2]**:** $O(V + E)$
- **Returns:** The sum of the weights of the longest path in the DAG.

[1] Finding the longest path in other types of graphs is at least NP-hard.

[2] This is a relatively straight-forward DP on a DAG problem if you wish to derive it.

# Graph Problems: General Stuff

**How to solve graph problems:**

1. Identify type of problem (Reachability, Shortest Path, [redacted])
2. Construct new graph
   - Add sources/sinks
   - Add vertices via $V' = V \times \{\text{some set}\}$ (Useful for tracking states)
   - Add ~~vertices~~ via $E' = E \times \{\text{some set}\}$ (Useful for allowing/prohibit certain behaviour)
3. Apply some stock algorithm **(DO NOT MODIFY THE ALGORITHMS - MODIFY THE INPUTS!)**
4. Draw connection between how to result of the algorithm upon the new graph relates to the solution of the original question.

# Recurrences and Asymptotics

Give a tight asymptotic bound for each recurrence:

- $T(n) = 4\ T(\frac{n}{2}) + n \log_2 n$
- $T(n) = T(\frac{3n}{4}) + T(\frac{n}{4}) + 5n$
- $T(n) = 9\ T(\frac{n}{3}) + n^2$

Group the following functions s.t. $f$ and $g$ are in the same group if $f(x) \sim \Theta(g(x))$, and sort the groups by runtime:

- $n^n$
- $\log \log n$
- $374^n$
- $n!$
- $\log(n + n^{374})$
- $\log n^n$
- $n^{1.000001}$
- $2^n$
- $n \log n^5$
- $\frac{1}{\log_n 2}$

# Recurrences and Asymptotics

# Divide and Conquer

Suppose you have two arrays: $A$ of length $n_1$, sorted in ascending order, and $B$ of length $n_2$, sorted in descending order. You decide to concatenate them to form a new array $C = A \cdot B$ for some application.

All the numbers are distinct. After concatenation, you no longer care about $A$ and $B$ individually, and you don't know $n_1$ or $n_2$. You only have $C$ and its size $n = n_1 + n_2$.

For example, if $A = [1, 2, 6, 7]$ and $B = [8, 4, 3]$, then $C = [1, 2, 6, 7, 8, 4, 3]$.

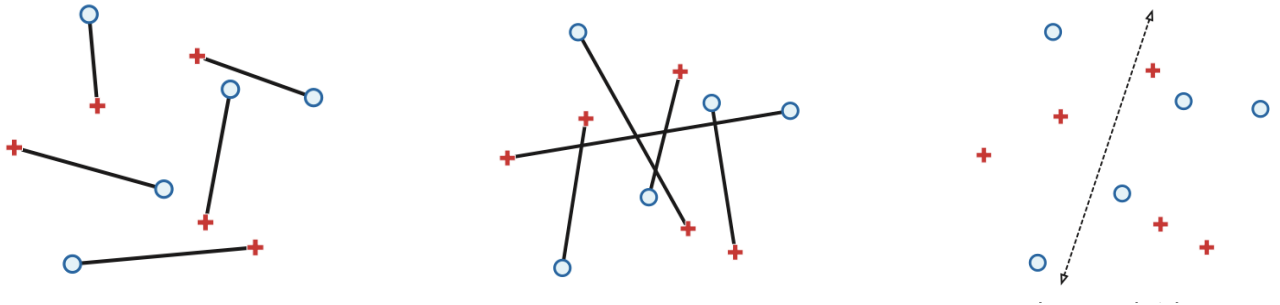
Given a number $x$, we want to check whether $x$ appears in $C$.

# Divide and Conquer

# Divide and Conquer

Given sets $R, B \subset \mathbb{R}^2$ ($n$ red points and $n$ blue points) in the plane, we define

- **Tidy Matching:** *n non-intersecting* segments pairing each red point to a unique blue point. (Fig. 1(a) vs 1(b))
- **Ham-sandwich cut $\ell$:** A line with exactly $\lfloor n/2 \rfloor$ red and $\lfloor n/2 \rfloor$ blue points strictly above it. (Fig. 1(c))



(a) Tidy Matching (b) Not a tidy matching (c) Ham-Sandwich Cut

*Fact: A valid tidy matching and ham-sandwich cut always exist.*

**Black-Box Subroutines:**

- `HamSammy(R, B)` : Returns a ham-sandwich cut $\ell$ in $O(n)$ time.
- `Above?(p, ℓ)` : Returns `True` if $p$ is strictly above $\ell$ in $O(1)$ time.

Describe and analyze an algorithm to compute a Tidy Matching of $R$ and $B$, using these subroutines.

# Divide and Conquer

# Dynamic Programming

*A shuffle* of two strings $X$ and $Y$ is formed by interspersing the characters into a new string, keeping the characters of $X$ and $Y$ in the same order. For example, the string BANANAANANAS is a shuffle of the strings BANANA and ANANAS in several different ways.

BANANAANANAS BANANAANANAS BANANAANANAS

Similarly, the strings PRODGYRNAMAMMIINCG and DYPRONGARMAMMICING are both shuffles of the strings DYNAMIC and PROGRAMMING:

PRODGYRNAMAMMIINCG DYPRONGARMAMMICING

Given three strings $A[1..m]$, $B[1..n]$, and $C[1..m+n]$, describe and analyze an algorithm to determine whether $C$ is a shuffle of $A$ and $B$.

# DP

$A[1..n]$, $B[1..m]$ $C[1..n+m]$

$(i)$ $(i+j)$

1.) English description:

Let IsShuffle(i, j) be 0 when $A[i..n]$, $B[j..m]$ is a shuffle of $C[i+j..n+m]$ and not 0 otherwise

2.) $IsShuffle(i,j) =$

$$\begin{cases} 0 & i+j > n+m \\ IsShuffle(i,j+1) & i > n \\ 1 \text{ if } B[j] = C[i+j] & \\ IsShuffle(i+1,j) \text{ if } A[i] = C[i+j] & \\ 0 \text{ if not} & \end{cases}$$

3.) We return $IsShuffle(1,1)$

$\Rightarrow$ $\min\{IsShuffle(i+1,j) \text{ if } A[i] == C[i+j],\ IsShuffle(i,j+1) \text{ if } B[j] == C[i+j]\}$

4.) Memoize in $IsShuffle[1..n+1][1..m+1]$

5.) Fill $IsShuffle[i][j]$ in decreasing $i+j$ order



# What *is* it about cephalopods?

An entrepreneurial octopus runs a "Fish Wash" at the local coral reef. At exactly 8:00 AM, a morning rush of $n$ fish arrive to be cleaned.

To handle the rush, the octopus acts as a multi-lane car wash, dedicating exactly **3 of its arms** to scrub the fish.

Each fish $i$ (for $1 \leq i \leq n$) requires $\ell_i$ minutes to scrub clean, where $\ell_i$ is a positive integer at most $L$. The octopus assigns each fish to one of the 3 arms. Fish assigned to the same arm form a line and must be cleaned in their original sequential order.

Describe an algorithm to assign the fish to the 3 arms in a way that **minimizes the sum of the waiting times** for all the fish. Analyze the runtime of your algorithm in terms of $n$ and $L$.

---

**Example:** $\ell = [5, 3, 10, 20, 6, 15]$ and $L = 20$. Suppose the octopus assigns fish 1 and 3 to Arm 1; fish 2, 4, and 5 to Arm 2; and fish 6 to Arm 3.

- Fish 3 waits 5 mins (while Arm 1 cleans Fish 1).
- Fish 4 waits 3 mins (while Arm 2 cleans Fish 2).
- Fish 5 waits $3 + 20 = 23$ mins (while Arm 2 cleans Fish 2 and 4).

The total waiting time is $5 + 3 + 23 = 31$ minutes.

# DP

# Dynamic Programming

The **StupidScript** language includes a binary operator @ that computes the *average* of its two arguments. For example, the StupidScript code `print(3 @ 6)` would print 4.5, because (3 + 6)/2 = 4.5.
Expressions like 3@7@4 that use the @ operator more than once yield different results when they are evaluated in different orders:

$$(3@7)@4 = 5@4 = 4.5 \quad \text{but} \quad 3@(7@4) = 3@5.5 = 4.25$$

Here is a larger example:

$$\begin{aligned}(((((8@6)@7)@5)@3)@(0@9)) &= 4.5\\ ((8@6)@(7@5))@((3@0)@9) &= 5.875\\ (8@(6@(7@(5@(3@0)))))@9 &= 7.890625\end{aligned}$$

Your goal for this problem is to describe and analyze an algorithm to compute, given a sequence of integers separated by @ signs, the **largest possible** value the expression can take by adding parentheses. Your input is an array $A[1 \ldots n]$ listing the sequence of integers.

For example, if your input sequence is $[3, 7, 4]$, your algorithm should return 4.5, and if your input sequence is $[8, 6, 7, 5, 3, 0, 9]$, your algorithm should return 7.890625. Assume all arithmetic operations (including @) can be performed exactly in $O(1)$ time.

# DP

# Bad Apple

Jeff is falling into a Bad Apple rabbit hole on Youtube. He wants to get from the original **Bad Apple** video to the video **bad apple but it's a sorting algorithm**[3] by *only* watching videos that are recommended to him in sequence.

Formally, you are given a list of $n$ videos $V$ such that each $v \in V$ has a watch time $\tau(v)$ and list $R(v) \subseteq V$ of sidebar recommended videos that can be watched after $v$ is watched. Suppose video $s \in V$ is the original **Bad Apple** video and $t \in V$ is the **bad apple but it's a sorting algorithm** video. We call a sequence of videos starting from $s$ and ending at $t$ a *Terpsichorean marathon*.

(a) Describe and analyze an algorithm to find a *Terpsichorean marathon* of minimum total watch time.

(b) We call a video a *banger* if it is in some *Terpsichorean marathon* with minimum watch time. Design and analyze an efficient algorithm to find all the *banger* videos.

a.) Let $G = (V, E)$ be a directed and weighted graph.

$V' = V \cup \{s'\}$

$E' = \{u \to v \mid u \in V \text{ and } R(u) = v\} \cup \{s' \to s\}$

Let $w(u \to v) = \tau(v)$

[3] Yes, this is a real thing.

$O(V') = O(V) = O(n)$. $O(E') \leq O(V^2) = O(n^2)$

## Graphs

The shortest s-t walk of minimum weight is the shortest path from s' to t' in G'.

Run Dijkstra's from s' to t.

Reconstruct the path from Dijkstra's in $O(V'+E')$ time and return the path

Take $O(E' \log V') = O(n^2 \log n)$ time

4.) Run parts 3a) return the minimum length and place it in len. Construct the whole graph $G' = (V'', E'')$

$V'' = V'$

$E'' = \{u \to v \mid v \to u \in E'\}$

Run Dijkstra's on G from s' and store in dist

Run Dijkstra's on G' from t and store in dist$^R$

# Who's the number one spice?

Young Cardamom is bringing the flavor to the fish![4] His dish needs to have both **HEAT** and **SWEET** spices to be complete.

The kitchen can be modeled as a DAG where:

- Vertices represent prep stations
- Each station is labeled either HEAT, SWEET, or NEUTRAL
- A directed edge $(u, v)$ with weight $f(u, v)$ means station $u$ can send ingredients to station $v$, contributing flavor value $f(u, v)$
- Young Cardamom starts at station $s$ and must reach plating station $t$

To achieve the #1 spice, Young Cardamom's path must visit **at least one HEAT station** and **at least one SWEET station**.

Design and analyze an algorithm to find a path from $s$ to $t$ that visits both required station types and maximizes the total flavor value.

[4] Of course, HAB is bringing the flavor to the rice.

# Graphs

# Graphs

We are given a directed graph $G = (V, E)$ with $n$ vertices and $m$ edges, where $m \geq n$. Each edge $e \in E$ has a weight $w(e) \geq 0$, and all weights are distinct. Let $s$ and $t$ be two distinct vertices in $V$.
For a path $\pi$ from $s$ to $t$, we define its **bottleneck value** as

$$b(\pi) = \max_{e \in \pi} w(e),$$

i.e., the weight of the heaviest edge along $\pi$.
We want to find a path from $s$ to $t$ that **minimizes the bottleneck value**.

# Graphs

# Feedback

- Further questions in ACM (Siebel 1104) after this
- Please fill out the feedback form:
  `go.acm.illinois.edu/cs374a_mt2_feedback`

